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HILLSLOPE RUNOFF PROCESSES AND MODELS

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ABSTRACT

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Hillslope hydrology is concerned with the partition of precipitation as it passes through the vegetation and soil between overland flow and subsurface flow. Flow follows routes which attenuate and delay the flow to different extents, so that a knowledge of the relevant mechanisms is important. In the 1960s and 1970s, hillslope hydrology developed as a distinct topic through the application of new field observations to develop a generation of physically based forecasting models. In its short history, theory has continually been overturned by field observation. Thus the current tendency, particularly among temperate zone hydrologists, to dismiss all Hortonian overland flow as a myth, is now being corrected by a number of significant field studies which reveal the great range in both climatic and hillslope conditions.

Some recent models have generally attempted to simplify the processes acting, for example including only vertical unsaturated flow and lateral saturated flows. Others explicitly forecast partial or contributing areas. With hindsight, the most complete and distributed models have generally shown little forecasting advantage over simpler approaches, perhaps trending towards reliable models which can run on desk top microcomputers. The variety now being recognised in hillslope hydrological responses should also lead to models which take account of more complex interactions, even if initially with a less secure physical and mathematical basis than the Richards equation. In particular, there is a need to respond to the variety of climatic responses, and to spatial variability on and beneath the surface, including the role of seepage macropores and pipes which call into question whether the hillside can be treated as a Darcian flow system.

THE SCOPE AND RELEVANCE OF HILLSLOPE HYDROLOGY

More than 95% of the water in streamflow has passed over or through a hillside and its soils before reaching the channel network. Hillslope hydrology is concerned with the transformation of precipitation as it passes through the vegetation and soil layers. Much water is lost through transpiration and evaporation of both intercepted water and soil water, and some water percolates through the soil zone to contribute to groundwater at depth. The present review is concerned primarily with flow processes within the soil and over its surface. Although water quality, particularly for solutes, is also transformed within the soil, this topic is excluded from discussion here.

Net precipitation is partitioned at the surface between *overland flow* and *subsurface flow*. No distinction can be drawn in principle between subsurface flow within the soil and groundwater flow but in practice flows in soil and/or

regolith rather than in bedrock are of concern. *Hortonian overland flow* is that produced when rainfall or snowmelt rates exceed the current infiltration capacity of the soil. *Saturation overland flow* is produced when the storage capacity of the soil is completely filled, so that all subsequent additions of water at the surface, irrespective of their rate of application, are forced to flow over the surface. *Return flow* occurs when subsurface flow is constrained to flow out of the soil, overland, in areas of profile concavity and/or flow convergence in plan, or where soil thickness and/or permeability are decreasing downslope. These types of flow follow routes which attenuate and delay the flow to different extents (Fig. 1), so that a knowledge of the relevant mechanisms is an important first step in understanding hillslope hydrology within the context of the catchment.

Movement of soil water can, in many cases, be described as a Darcian flow, with only modest dispersion of a sharp front between wet and dry soil, or between successive identifiable slugs of water. Darcy's law breaks down where there are continuously connected large voids (macropores and pipes) which allow significant bypassing of the main flow which may be turbulent. Another

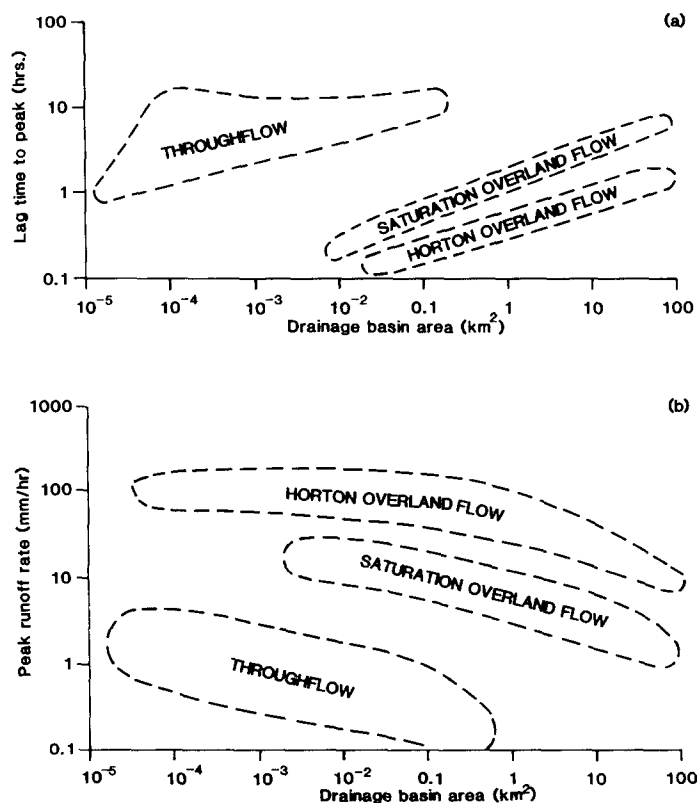


Fig. 1. Generalised response of catchments to hillslope flow processes (after Dunne, 1978). (a) Lag times; (b) Peak runoff rates.

problem is that the non-capillary voids may behave as a dendritic network, carrying a high proportion of the total hillslope flow towards a relatively small number of discrete seepage lines, which may or may not be recognised as visible pipes. Hillslope response may then reflect the properties of this, largely unseen, dynamic network.

HISTORICAL CONTEXT

Although there are previous scattered references to hillslope hydrological processes, Horton's (1933) infiltration theory was the received view, at least until the 1960s. A notable exception was the work of Hursh (e.g. Hursh and Fletcher, 1942). The conflict in humid areas between the presence of "surface runoff" in the stream hydrograph with an evident absence of Hortonian overland flow was first recognised widely for undisturbed forest soils (e.g., Vasilyev, 1948; Van Dijk, 1958). Hewlett (Hewlett, 1961; Hewlett and Hibbert, 1963) played a crucial role in putting this work on a sound footing and demonstrated that low flows could, in suitable areas, be maintained entirely by subsurface flow within the soil. Whipkey (1965) then showed experimentally that flood peaks might be largely or completely generated from subsurface flow in suitable soils. Betson (1964) demonstrated that overland flow from only a small proportion of the catchment area was, in many cases, sufficient to supply the flood hydrograph peak. Although the "partial area" or "contributing area" concept has been used in several distinct forms, the essential concepts of current hillslope hydrology had been introduced by the mid-1960s and were swiftly tested in other areas (e.g., Ragan, 1968; Dunne and Black, 1970a,b; Weyman, 1970). Low and high flows could, in suitable circumstances, be produced by subsurface flow; and high flows could be produced by overland flow from only a small fraction of the catchment area.

The incorporation of these central concepts into forecasting models is still in progress. The tendency, particularly among temperate zone hydrologists, to dismiss all Hortonian overland flow as a myth, has been corrected as a number of significant field studies reveal the great variety of possible hillslope responses. The development of forecasting models, with an explicit hillslope component advanced most quickly for Hortonian overland flow, before alternative flow concepts had been generally accepted. Wooding (1965a,b) and Woolhiser (1969) analysed the conditions under which the kinematic wave solution was appropriate, for simplified catchment shapes with uniform production of overland flow. Smith and Woolhiser (1971a,b) and Freeze (Freeze, 1972a,b; Stephenson and Freeze, 1974) developed a similarly rigorous treatment of saturated and unsaturated subsurface flow which showed both the strengths and limits of fully distributed models. This work clarified the relatively narrow range of conditions under which return flow could occur, as a means of establishing a dynamic area contributing saturation overland flow. It also showed the very high intensity of field data needed to provide adequate parameterization for a fully distributed model. Subsequent numerical models have tended

to use less complete specifications of the hillslopes. For example, Hewlett and Troendle (1975), subdivided the hillslope into schematic elements to forecast the saturated contributing area at the heart of their model. Beven and Kirkby (1979) took this process still further by selecting a simplified mathematical form for the subsurface flow to provide an explicit solution for (areally uniform) subsurface runoff, and hence to forecast contributing area. Similar compromises are used in most current models (e.g. the SHE model, described by Beven et al., 1980). Full distribution of parameters occurs typically only at the subcatchment level, where the potential of remote sensing methods for data acquisition appears most promising (e.g. Huff et al., 1982).

Perhaps the most significant growth area in hillslope hydrology is in the development of variety, led by continuing field studies. Macropores (e.g., Beven and Germann, 1981; Smettem, 1984) and pipes (e.g. Gilman and Newson, 1980; Jones, 1971) added to problems of textural heterogeneity, show the need for questioning the Darcian assumptions for subsurface flow. Evidence from arid areas (e.g., Yair and Lavee, 1985; Dunne and Aubry, 1986) re-emphasises the importance of overland flow, and recalls Emmett's (1970) field experiments on the lateral distribution of flow depth and roughness. Furthermore, overland flow need not persist long enough to accumulate down the full length of a slope but may re-infiltrate between rain showers. At another climatic extreme, Woo and Steer (1982, 1983) show the effect of active layer thawing on overland flow in the Arctic. The combined thrust of this work is to show the great range in both climatic and hillslope conditions, which lead to a corresponding variety in hillslope hydrological responses.

HILLSLOPE RUNOFF PROCESSES

At any point on a hillslope surface, or within the soil, water is moving under potential gradients due to both gravity and water pressure or tension. For practical purposes, water may be treated as incompressible, and continuity of mass or volume must be maintained, taking into account inputs and outputs, including vapour transfers. Flow through fine pores in the soil matrix is resisted mainly by viscous stresses. Where pore sizes are randomly interconnected, the aggregate flow may be described satisfactorily by Darcy's law. For flow through large voids and overland, the critical Reynolds number for the onset of turbulence may be exceeded, so that inertial resistances predominate, and it may also be necessary to take flow bypassing into account.

For the Darcian case, two-dimensional flow conforms to the continuity equation:

$$\nabla Q + \partial m / \partial t = a \quad (1)$$

where: Q is the vector discharge per unit cross-section; m is the volumetric moisture fraction; t is elapsed time; and a is the rate of addition of water (normally zero). For a non-isotropic soil, discharge due to a total potential field Φ is:

$$Q = -K_x \partial \Phi / \partial x \mathbf{i} - K_y \partial \Phi / \partial y \mathbf{j} \quad (2)$$

where K_x , K_y are hydraulic conductivities, which depend strongly on moisture content; and $\mathbf{i}$, $\mathbf{j}$ are unit vectors in the x (horizontal down-flow), and y (vertical downwards) directions.

These equations may be solved completely, using numerical methods and suitable expressions or tables of values for hydraulic conductivity in terms of soil moisture (e.g. Hahn et al., 1982). Hydraulic conductivity also depends on moisture history. In a pore system of non-uniform cross section, hydraulic potential is controlled by the narrowest necks but moisture content in the broadest sections depends on how much they have been filled previously. The relationship between conductivity and moisture content therefore follows different curves during wetting and drying cycles, although numerical solutions are often restricted to conditions of progressive wetting up, or else the hysteresis is ignored. For the isotropic case, ($K_x = K_y = K$), eqn. (2) is generally rewritten as the Richards (1931) equation in the form:

$$Q = -K \nabla \cdot \Phi = K \mathbf{i} - K \nabla m \partial \Psi / \partial m = K \mathbf{i} - D \nabla \cdot \Phi \quad (3)$$

where: the unit vector $\mathbf{i}$ is in the vertical downward (z) direction; Ψ is the hydraulic potential ($= \Phi + z$); m is the volumetric moisture fraction; and the hydraulic diffusivity D is defined as $D = K \partial \Psi / \partial m$, which is treated as a second, related empirical function of moisture content.

The analysis of hillslope flow may be simplified by noting that the direction of flow is commonly either nearly vertical or nearly downslope. For a hillside with uniform net rainfall, flow vectors are typically vertical through the soil, however the exceptions to vertical flow are critical to hillslope hydrology, because they produce substantial diversions of flow towards the downslope direction. Lateral flow diversion may occur: (1) in a zone where saturated vertical discharge falls below the net rate of percolation from above; (2) where percolating water reaches a perched or general zone of saturation; or (3) where lateral hydraulic conductivity greatly exceeds vertical conductivity.

Within the soil, vertical conductivity generally declines downwards into the soil, although there are many exceptions to this rule, and changes need not be gradual. Typically the increase in overburden and the decline in bioturbation reduce porosity with depth, but soils may show sedimentation and/or weathering features which reverse this general trend, and some parent materials show a greater permeability than the regolith on them. In a zone where vertical conductivity decreases downwards, moisture content tends to rise until, if percolation is rapid enough, saturation is reached. A saturated layer then builds up as a perched water table. Beneath either a perched or a regional water table, flow is constrained by the continuity equation. If hydraulic conductivity remains high at depth, as occurs over a good aquifer, there may be a substantial vertical flow component, but, more commonly, the decline of conductivity with depth concentrates flow laterally and close to the water table surface. Comparable mechanisms of flow diversion occur at the

surface, giving rise to Hortonian or saturation overland flow, even though overland flow is not Darcian.

In the unsaturated zone, flow vectors are normally close to the vertical, so that downslope components are usually relatively slight for conditions of overall precipitation. Lateral conductivity must be many times higher than vertical to give significant lateral flow. Major causes of anisotropy are root holes, faunal burrows and "thatching" by surface leaf litter, but under unsaturated conditions, the associated voids carry very little flow. The most important role of anisotropy in the unsaturated zone is therefore in providing macropores which allow vertical bypassing, so that some precipitation reaches a saturated zone much more quickly than through unsaturated pores and produces a rapid lateral response from the saturated zone.

Flow processes are described below on the basis of the commonest combinations of process and flow directions but for cases which are not covered in this way, the full equations above must be solved numerically, in either a finite element or a finite difference scheme, such as that used by Freeze (1972a, b). The simple cases described are infiltration, saturated lateral subsurface flow, macropore and pipe flow, and overland flow. Conditions near the stream bank, where hydraulic potential surfaces commonly intersect the surface steeply, provide the most important exception.

INFILTRATION INTO AND THROUGH THE SOIL

Infiltration equations are concerned with the rate at which water can enter the soil and are widely used to estimate the rate of production of Hortonian overland flow. On real surfaces microtopography and soil heterogeneity can greatly modify the mean rates of infiltration. Within the soil, the rate of percolation through the unsaturated zone has a critical influence on the form and timing of the hillslope hydrograph. This response can be very different for soils with mainly Darcian flow and in soils with significant macropore bypassing.

The most widely used expression for infiltration rate is that due to Philip (1957):

$$f = A + (B/2t)^{1/2} \quad (4)$$

where: f is the maximum infiltration rate, or infiltration capacity, at time t after water is impounded at the soil surface; and A , B are constants for a given soil and given initial conditions. This expression may be derived from the one-dimensional form of eqn. (3) above, for fixed K and D , and it may be seen that $A = K$ and that $B = 2D^2$. During observations of infiltration, a rather sharp wetting front commonly advances down into the dry soil, with near-saturation behind it. The constants in eqn. (4) may therefore be roughly identified with the saturated hydraulic conductivity and with the diffusivity at the wetting front. Philip has shown that these terms are a valid first approximation and gives a more exact identification for A and B .

A second, older approach which has been usefully combined with Philip's work is the Green and Ampt (1911) equation, which expresses infiltration in terms of a conceptual soil water store, S . The infiltration capacity is then:

$$f = A + B/S \quad (5)$$

If the conceptual store is considered to receive rainfall, and to leak at a constant rate A , then eqns. (4) and (5) are equivalent for ponding infiltration, with the same parameter values A and B . The strength of the storage approach is that it is able to provide a simple forecast for infiltration rates during non-ponding conditions. It may, for example, give a rational forecast for the onset of ponding conditions during a steady rainfall at intensity i :

$$t_p = B/(i - A)^2 \quad (6)$$

The storage approach also allows for the possibility of saturation of the conceptual store. For a total capacity S_c , the ponding time is then:

$$t_p = S_c/(i - A) \quad (7)$$

It may readily be seen that at high intensities ponding occurs through exceeding the infiltration capacity and at low intensities through exceeding the saturation capacity.

On natural surfaces, there is an opportunity for infiltration at every point while rain is actually falling. Once rain has stopped, surface irregularities channel any overland flow and ponds depression storage near the lowest points of the surface, where infiltration continues until surface water is exhausted. Dunne (Dunne and Dietrich, 1980; Dunne and Aubry, 1986) and his co-workers have shown that, on sparsely vegetated surfaces in Kenya, topographic lows have systematically lower infiltration capacity than highs. This occurs through sedimentation and infilling of large pores and burrows and because vegetation and organic matter tend to occur on mounds. As a result, average infiltration rates after rainfall increase more than linearly with the proportion of the area inundated. If overland flow runoff increases downslope, then there is a consistent downslope increase in total infiltration through this mechanism.

On rocky surfaces, infiltration is confined to cracks, which may be filled with inwashed fine material. Within cracks, water penetrates deeply and supports vegetation within otherwise barren areas. The effect is evident in the Negev Highlands of Israel (Yair, 1983) in an area of less than 200 mm annual rainfall, and is commonly found in other rocky areas (including karst). Since cracks generally form topographic lows in the surface, average infiltration rate tends to increase less than linearly with proportion of the surface inundated in this case. The average infiltration rate in rocky areas is generally low, producing overland flow which can infiltrate more freely along downslope soil or sediment margins. These examples show that infiltration, particularly after rainfall has ceased, shows strong areal effects which depend on surface microtopography. These effects influence average infiltration amounts and introduce considerable spatial heterogeneity in the depth of penetration of the infiltrated water.

There are no simple expressions, like those for infiltration capacity, to forecast percolation within the soil. If Darcian flow can be assumed, as it can in many cases, then there is no difficulty in principle in computing one-dimensional numerical solutions to the Richards and continuity equations, (3) and (1) above. This approach is however relatively demanding of data and to some extent of computer time. Where cracks, faunal burrows or other macropores carry a significant fraction of the percolating water, the Darcian assumptions break down. There may then be significant amounts of flow bypassing and great heterogeneity in percolation depths. These effects may be associated with the patterns of surface infiltration illustrated above.

Under conditions of Darcian flow, textural voids convey the percolating water as a plug flow with only moderate dispersion. Infiltrating water then commonly penetrates at a moisture content close to saturation, with a relatively sharp wetting front below. Where there is a saturated layer at shallow depth, infiltrating water is commonly superimposed on an antecedent profile of increasing soil moisture with depth, developed in response to downward drainage balanced by upward diffusive flow and evapotranspiration. After the end of rainfall, the surface soil layers drain while the wetting front advances downward and becomes less pronounced. The rate of percolation into the top of the saturated layer shows a strongly non-linear response to both the depth of the wetting front and the amount of storm rainfall, as is illustrated in Fig. 2 for two storms of different duration. During the longer storm (Fig. 2b), the wetting front is pushed closer to the saturated layer, and, for storms over about 5 h, the two finally coalesce. After rainfall has ceased, the longer storm has penetrated to wetter, though still unsaturated, subsurface layers. Percolation therefore reaches the saturated layer sooner and at a higher rate. This is one major source of the non-linearity observed in hillslope hydrological response.

A soil with large, continuous macropores shows behaviour at the other extreme from the strictly Darcian soil described above. If macropores are large, but relatively few, then rainfall in excess of the textural infiltration capacity is funnelled down the macropores at a rate which increases over time, as the textural infiltration capacity falls. This water flows down the macropore walls as a thin film, soaking into the walls as it goes. Its rate of advance is controlled by the lateral infiltration rate, if the soil is initially dry, and by flow resistance (initially laminar but eventually turbulent), if the macropore walls are initially wet. For the dry wall case, lateral infiltration is most rapid at the advancing flow front, which is consequently abrupt. Figure 3a illustrates the calculated advance of flow down the macropore, and infiltration into the walls, for a constant inflow from the surface in a film 0.1 mm thick. At depth z and time t after inflow at rate q_0 has started, the cumulative lateral infiltration is given by:

$$F = B [2t/B - (\pi z/q_0)^2]^{1/2} \quad (8)$$

and the discharge down the pore wall by:

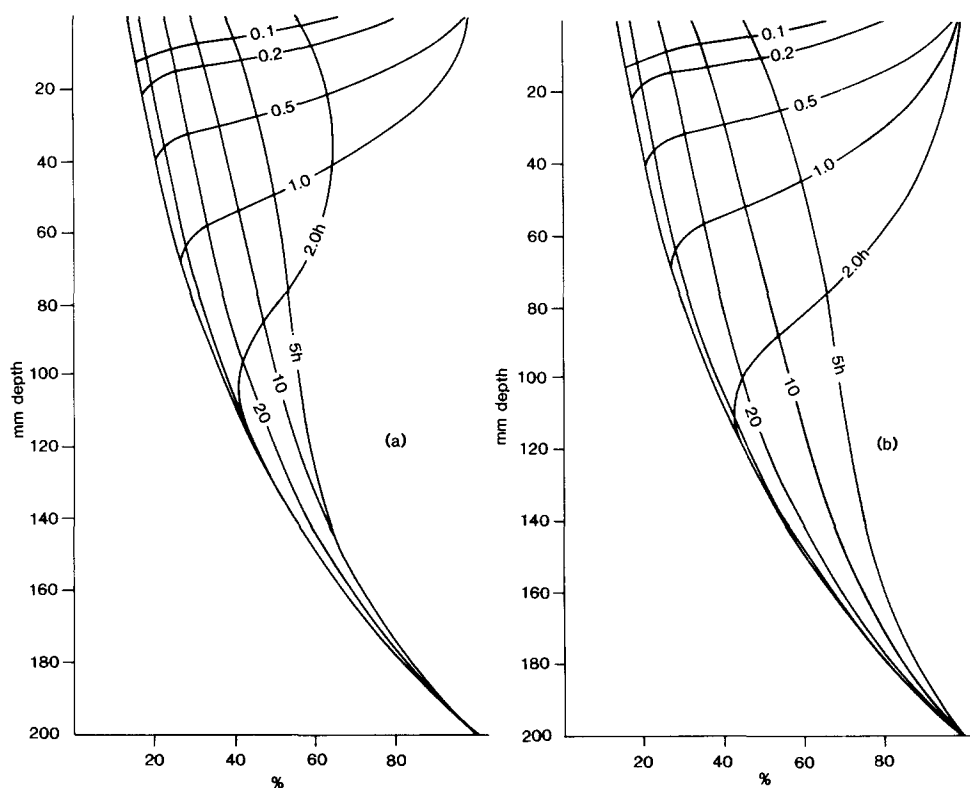


Fig. 2. Simulated moisture profiles during infiltration of rainfall at 20 mm h^{-1} , and percolation down to a saturated layer at 200 mm. Moisture conditions prior to rainfall in equilibrium with saturated layer. (a) Rainfall for 1 h; (b) Rainfall for 2 h.

$$q = q_0 \{1 - 2/\pi \sin^{-1}[\pi z/q_0 (B/2t)^{1/2}]\} \quad (9)$$

In the example shown in Fig. 3a, lateral infiltration is at the assumed rate $B = 500 \text{ mm h}^{-1}$, corresponding to capillary flow in linear textural pores of width $0.072 \mu\text{m}$. It can be seen that 100 mm of water has infiltrated laterally over a 10 h period, with penetration of 250 mm assuming $\varepsilon = 40\%$ porosity. In the same period, flow down the macropore has penetrated 750 mm, so that the macropore flow is significantly bypassing the textural pores. For a soil with this texture, the critical macropore width to sustain enough flow to produce bypassing is $140 \mu\text{m}$; in general, the critical width is:

$$r_c = 2 [3\mu\pi B/g\varepsilon]^{1/3} \quad (10)$$

Figure 3b shows calculated critical macropore diameters corresponding to the effective textural pore diameters responsible for the infiltration parameter B . The extent of bypassing by macropores is related to the degree of bimodality in the population of soil pores. In the extreme case of cracking clay soils, bypassing may be considerable and widespread, whereas in coarse textured

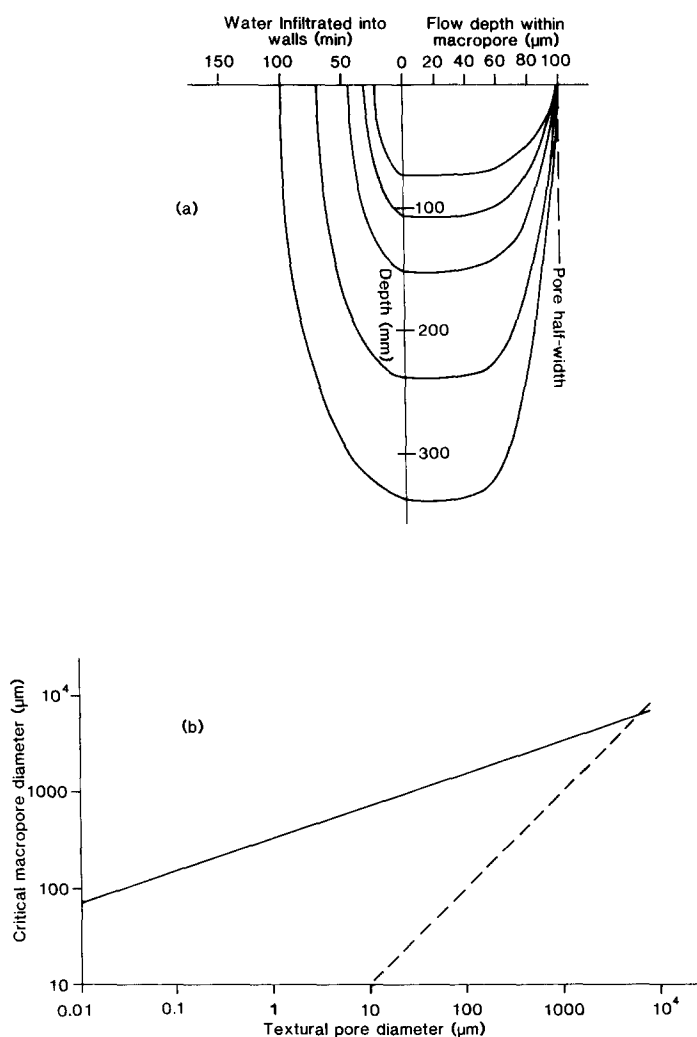


Fig. 3. Simulated flow down one side of a planar macropore, of width 0.2 mm, with lateral infiltration into an initially dry soil through the macropore wall. (a) mm of water infiltrated into macropore wall (at left) and thickness of flowing water film (at right); (b) Minimum macropore diameter needed to allow bypassing, expressed in terms of effective textural pore diameter (which controls lateral infiltration rate). Circle shows critical diameter for case shown in (a).

soils, bypassing may only be realised down faunal burrows and decayed root holes at exceptional rainfall intensities.

Where macropores penetrate previously wetted soil, then flow velocities reflect flow dynamics, with estimated velocities of 100 m h^{-1} for the 0.2 mm crack shown in Fig. 3. Where macropores are closely spaced, lateral infiltration will completely wet the soil between the macropores. In this case the flow may once more be considered as Darcian, though with parameters reflecting the macropore population rather than the textural pores alone. In practice, soils

contain a distribution of pore sizes, with varying degrees of continuity. Large, sparse macropores lead to significant bypassing of the textural infiltration and to a wetting front which is highly irregular in section. Under these circumstances, some rainfall is able to reach saturated layers much more rapidly than through textural pores alone, so that a hillslope hydrograph shows an early peak, with a response time which would usually be associated with an overland flow rather than subsurface flow. This is particularly evident where macropores through the unsaturated zone connect with natural soil pipes which carry water downslope at channel flow velocities.

Infiltration is a crucial process for shaping the hillslope hydrograph, it introduces delays ranging from a few minutes to several days before there is any significant response from the saturated zone. Although Darcian flow is well understood, there is a need for considerable work on the effects of microtopography and other spatial heterogeneities and on the importance of macropores for individual sites. There is, at present, no unified body of theory which effectively addresses these issues.

SATURATED LATERAL SUBSURFACE FLOW

Once water has reached a saturated zone, whether it is a part of the regional water table, or a perched layer within the regolith, its behaviour is somewhat simpler than for the unsaturated zone. Downslope movement generally predominates, and is greatest near the top of the saturated zone. Where flow occurs in an interconnected set of voids, the flow can generally be treated as Darcian but, where continuous pipes occur, they can again produce significant bypassing of the Darcian flow and speed the hydrograph response.

For Darcian systems, the methods available to groundwater hydrology are generally applicable, and are reviewed elsewhere (e.g. Freeze and Cherry, 1979; Konikow and Patten, 1985; Konikow and Mercer, this volume). The main point at which near surface groundwater systems commonly differ from aquifer systems is that hydraulic conductivity may decrease rapidly with depth within the regolith. Where the underlying rock, although saturated, is essentially impermeable, the only effectively flowing layer is within the regolith. The effective transmissivity is then dependent on soil properties and may show great sensitivity to the level of flow, relative to the topographic surface. Hydraulic gradient may be assumed to parallel topographic gradient; and hydraulic conductivity, for uniform soils, may be taken as dependent on the local deficit below saturation, D . By integrating downwards, total transmissivity is also a decreasing function $\Phi(D)$ of saturation deficit, lateral flow is given by:

$$q = - \Phi(D) \partial z / \partial x \quad (11)$$

This may be combined with the one-dimensional continuity equation:

$$\partial(wq)/\partial x - w \partial D / \partial t = wi \quad (12)$$

where: x is the horizontal distance from the divide along a flow strip of local width w ; z is elevation; i is the rate of net rainfall; and t is elapsed time. Expressing discharge of q as aj , where a is the area drained per unit width at points on the flow strip and j is runoff per unit area, eqn. (12) can be re-expressed in the form:

$$a \partial j / \partial x - \partial D / \partial t = i - j \quad (13)$$

Equations (11) and (13) may be solved numerically for most plausible hillslope discharge functions, Φ . Of particular interest is the exponential function:

$$\Phi(D) = q_0 \exp(-D/M) \quad (14)$$

in which q_0 and M are soil parameters which are not allowed to vary over time, although q_0 may vary along the flow strip. They represent total saturated subsurface discharge on unit gradient (q_0) and subsurface saturated permeability (q_0/M), giving M an interpretation as a hydrologically effective soil depth. This expression is not only convenient mathematically, but is also a good approximation to the saturated subsurface response in many small humid temperate catchments. In this case, the two equations further reduce to:

$$a \partial j / \partial x + (M/j) \partial j / \partial t = i - j \quad (15)$$

This equation has the property that, if net input i is spatially uniform, the runoff j also becomes spatially uniform after a period of transient behaviour from an arbitrary initial state and thereafter remains uniform. The implication is that, to a first approximation, catchment runoff is uniform in amount and timing, so that the first term in eqn. (15) may be omitted; this gives a very simple runoff model for the flow strip:

$$M/j \, dj/dt = i - j \quad (16)$$

Local deficits may be calculated from local topography and soil parameters using eqn. (14). The assumptions of this approach are most closely met on convex divides, where discharge is proportional to gradient and deficits are everywhere similar. The conditions for spatially uniform input, from the unsaturated zone, are therefore most nearly met. On concave humid hillslopes, deficits tend to be progressively less at downslope sites, so that rainfall reaches the saturated zone earlier and earlier. In this case, eqn. (15) should be used in its full form. The approach breaks down where there is a perched saturated layer, giving a two-valued relationship between deficit and discharge; and in areas of low relief, where hydraulic gradient does not closely parallel the surface and changes significantly over time.

Equations (14) and (15) may also be used to show the approximate form of the subsurface saturated layer prior to a storm. For indefinite net rainfall at intensity i , runoff is equal to rainfall. Substituting in eqn. (14) for the equilibrium deficit:

$$D = M \ln(sq_0/ai) \quad (17)$$

where s has been written for the local slope, $-dz/dx$. The effect of topography and soil depth can be seen clearly. If solid depths (and therefore q_0) are assumed constant, then small deficits are associated with the base of concave slope profiles (low s) and hollows (areas of flow convergence in plan with high a). Small deficits are also seen to occur in areas of thin soils (low q_0) for example where resistant rock ribs approach the surface. As was noted above, divide convexities, on which gradient is nearly proportional to distance from the divide (i.e. a/s constant), show constant deficits. Since, in many humid landscapes, such convexities form a large part of the landscape, their simple hydrological response plays a large part in hillslope response. Equation (16) can also be used to show the general form of the hillslope (and stream) recession curve in a dry period, starting from an initial runoff j_0 . For $i = 0$, eqn. (16) then reduces to:

$$dj/dt = -j^2/M$$

with solution:

$$j = j_0/(1 + j_0 t/M)$$

$$D = M \ln(sq_0/aj_0) + M \ln(1 + j_0 t/M)$$

The relevance of this form may be seen by comparing the expression for runoff j with known stream recession curves, which show good agreement in many, though not all cases.

The analyses above are based on the assumption of Darcian flow in the saturated zone. Where piping is well developed, analyses based on the topographic index a/s are likely to be grossly in error (Jones, 1986). Pipes do not always run exactly orthogonal to contours, so that pipe catchment areas differ from surface catchments. Pipes also provide bypass routes for water in the saturated zone. To carry substantial flow, pipes must either originate well within the saturated zone, so that inflows occur under an appreciable pressure gradient (*seepage pipes*); or else the pipes must be fed through macropores carrying ponded water directly from the surface (*bypass pipes*). Much of the recent literature on pipes and their role in hillslope hydrology appears to be concerned with bypass pipes. Pipes may originate as animal burrows, reroofed channels or large cracks *inter alia*. They are frequently enlarged by hydraulic processes, which can erode new pipes up-flow from seepage sites and enlarge existing pipes, in some cases leading to collapse and the formation of an open gully.

Pipes vary in diameter from less than 10 mm to more than 2 m, the latter usually in semi-arid areas. Their size depends on their origins, on the stability of their walls and on their position relative to the water table, which may limit vertical development. Pipe networks carry water in turbulent flows, at velocities which match those for open channels, sometimes over distances of several hundred metres.

In a humid area, some pipes may carry perennial flow. In mid-Wales (Gilman, 1971; Jones, 1971, 1986), perennial pipes appear to drain areas of permanent

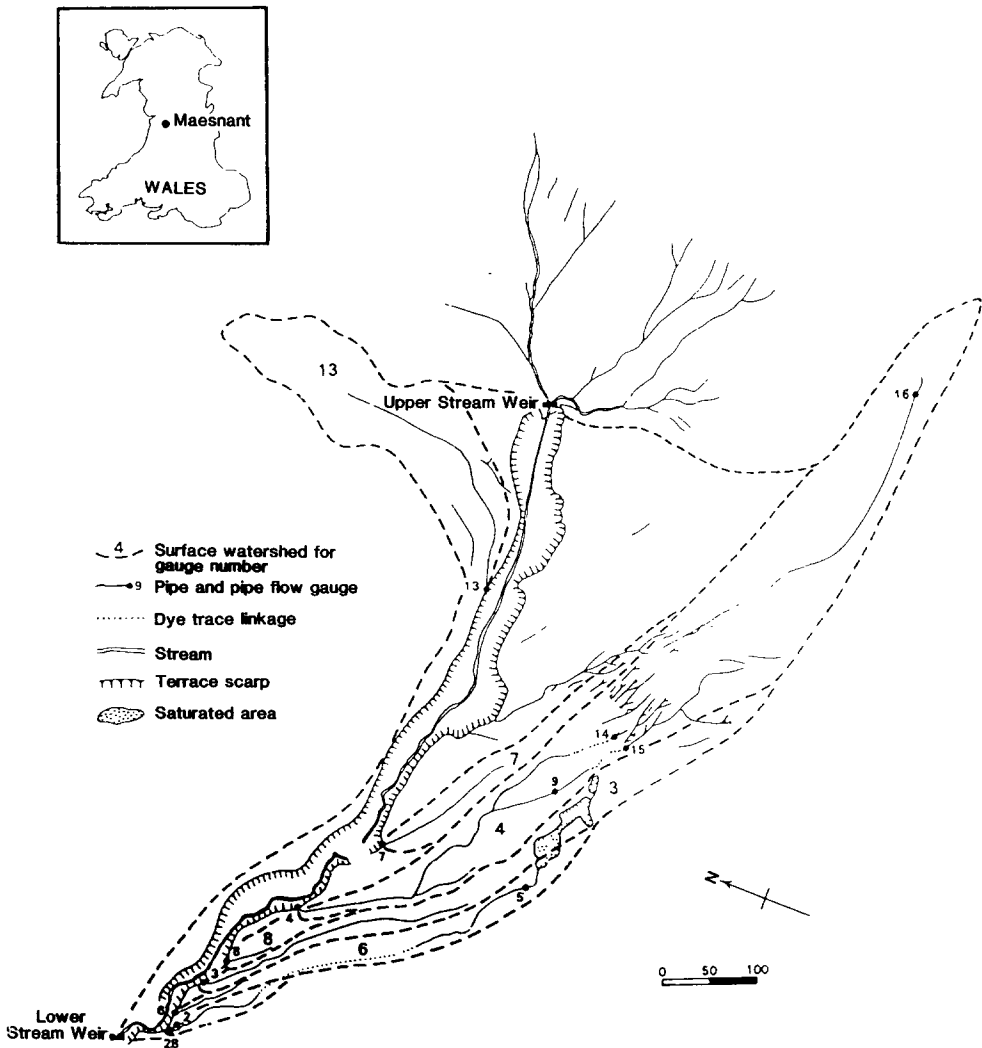


Fig. 4. Surface watersheds and traced pipe networks for the Maesnant experimental basin, in mid-Wales (from Jones, 1986).

surface saturation and so respond rapidly to storm rainfall. Ephemeral pipes in the area tend to be smaller and steeper than perennial pipes and to drain much smaller areas in storms. It is suggested that the ephemeral pipes have less opportunity for dynamic expansion of their channel and network through erosion. Where pipe networks have been mapped, requiring intensive field investigation (Fig. 4), it is possible to incorporate them explicitly into a hydrological forecast but this is not generally practiceable. To make forecasts, it may be necessary to consider flows generated by two extreme scenarios of no pipes and a maximum development of bypass pipes. For the former, it may be

assumed that all infiltration is Darcian, whether textural or through a dense network of macropores, supports diffuse subsurface flow; and for the latter that all infiltration directly supports either overland flow or pipe flow, which may be routed together to the slope base.

The role of seepage pipes is less clear, though they fulfil a similar role to artificial field drainage schemes, and much of the existing drainage theory may be applied to them. As for bypass pipes, their extent and density is usually unknown. It has been widely observed that measurements of subsurface flow, from open throughflow pits of 1 to 2 m width, generally indicate much less flow than measurements from longer pits or stream sections draining the same hillside (Dunne and Black, 1970a,b; Weyman, 1970; Parsons, 1987). Underestimation is commonly by a factor of 10 to 1000 times, although the timing of throughflow pit and stream hydrographs seems to be similar. Estimation of hillside hydraulic conductivity from streamflow similarly indicates a much higher value than that obtained from soil characteristics or from throughflow pits. It is inferred that networks of seepage pipes respond to fluctuations in saturation deficit in the same way as diffuse saturated subsurface flow. Their main effect may therefore be to redistribute the total outflow, with only rather slight effects on its timing. This has considerable implications for the effective hydraulic conductivities of hillslopes and may require that a simple Darcian view is modified. Some recent work (Binley et al., in press) shows that hillsides with random variations in hydraulic conductivity may not have a consistently meaningful mean conductivity, probably because of the development of preferential flow networks.

OVERLAND FLOW

Overland flow on real surfaces occurs as anastomosing flow with a great variety on flow depths within a small area. During rainfall on sparsely vegetated surfaces, the entire surface glistens with a thin water film. Mean velocities have been reported by Emmett (1970) in the range $1\text{--}30\text{ mm s}^{-1}$. His empirical exponent n in the relationship $v \propto r^n$, where v is the mean velocity and r is mean depth, takes values between 0 and 1. These values compare with theoretical exponents of 0.5 for laminar and 2.0 for turbulent flow. The actual flow regime does not conform well to either flow type, partly because shear resistance is distributed through the flow over projecting stones, vegetation stems and momentum exchange with raindrops; and partly because of the distribution of Reynolds numbers over the range of depths in any small area. Kirkby (1985) forecasts overland flow on the basis of a constant kinematic wave velocity (for a given surface) v_0 and a depth of depression storage r_0 which varies with slope gradient, in the form:

$$q = v_0 (r - r_0) \quad (18)$$

For Emmett's data appropriate values are $v_0 = 35\text{ mm s}^{-1}$, and $r_0 = 0.04\text{ mm}$ at gradient 0.33–1.4 mm at gradient 0.03. This gives the required range of depth exponents (n above) and the tendency for greater exponents on steeper slopes.

Overland flow in a thin layer is governed by the St. Venant equations for continuity:

$$1/w \partial(qw)/\partial x + \partial r/\partial t = i \quad (19)$$

and for motion:

$$\partial v/\partial t + v \partial v/\partial x + g \partial r/\partial x = g(s_0 - s_f) - vi/r \quad (20)$$

where: x is distance measured down a flow strip of width w ; q is overland flow discharge per unit width; r is mean flow depth; i is net input of rainfall less infiltration, etc; v is mean flow velocity, and s_0 , s_f are respectively the bed and friction slopes. These equations can be solved in their full form, but can, in many cases, be approximated well by neglecting the first two terms on the left-hand side of eqn. (20), giving the diffusion wave approximation; or the whole of the left hand side may be neglected to give the kinematic wave approximation:

$$s_0 = s_f.$$

The conditions under which various approximations are valid are reviewed by, among others, Morris and Woolhiser (1980). It is concluded that the kinematic approximation is widely applicable for thin overland flows so that the choice of a suitable friction slope becomes crucial. Solution have usually been obtained numerically, using the method of characteristics, for example by Wooding (1965a, b, 1966) and Woolhiser (Woolhiser and Liggett, 1967; Woolhiser, 1969; Kibler and Woolhiser, 1970), using turbulent flow relationships for the friction slope. For the formulation suggested above of constant kinematic wave velocity in eqn. (18) and, if the depression depth r_0 varies only slowly in the downslope direction, eqns. (19) and (20) reduce to the simplest form:

$$v_0 \partial(rw)/\partial x + \partial(rw)/\partial t = wi \quad (21)$$

Recent work on hillslope erosion, particularly that by Dunne (e.g., Dunne and Dietrich, 1980; Dunne and Aubry 1986) and his co-workers has re-emphasised the importance of correctly routing overland flow downslope, if appropriate erosion forecasts are to be made. The simplest case is where Hortonian overland flow is generated uniformly over the slope length and for long enough to allow flow to travel the full length of the slope during rainfall. In this case, total and peak flows are directly proportional to slope length. For the overland flow velocities quoted above, the time required for the flow wave to reach the base of a 100m slope is 48 min, whereas in arid areas (Yair and Lavee, 1985) individual rain showers may generally last only a few minutes. Except near the divides therefore, overland flow is almost constant downslope, or shows a modest increase with gradient, corresponding to higher mean velocities. If discharge is expressed as a power law of distance, then the exponent is well less than 1.0. In contrast, humid slopes typically generate saturated overland flow which, for normal convexo-concave slopes, is

negligible upslope and is produced at rapidly increasing rates per unit area near the slope base. If overland flow discharge, which accumulates these rates of production, is expressed as a power law of distance from the divide, its exponent for humid areas is therefore very much greater than 1.0. Thus there is a very wide range of response patterns, even over uniform soils, with implications for improved erosion forecasting as well as for hillslope hydrographs.

MODELLING HILLSLOPE RUNOFF

The unit for modelling hillslope response as a whole is plainly the flow-line strip, within which gradients and strip width may be allowed to vary, and which interacts negligibly with neighbouring strips. Not all hydrological models for catchments and large grid squares are built up from flow strips, but, if hillslope hydrological processes are recognised as important in shaping catchment hydrographs, then flow strips provide an efficient way of reducing the dimensionability of the solutions. Numerous models are primarily concerned with the individual hillslope processes discussed above. Here concern is directed towards models to integrate the processes into a synthesized hillslope hydrograph.

The most general models are those which attempt a full numerical solution of the governing equations, which are usually taken to be: (1) the two-dimensional Darcian flow equation [eqn. (2) above] for infiltration and subsurface flow, with values for conductivity and hydraulic potential taken from field measurements or a suitable generalised function; (2) the Saint Venant equation for overland flow, eqn. (20) above, with a turbulent flow friction law such as the Manning or Chezy equation; (3) appropriate continuity equations; and (4) boundary conditions defined by the flow divides, the base of the soil, stream outflow and rainfall at the soil surface. This approach has been pursued most successfully by Freeze (Freeze, 1972a,b; Stephenson and Freeze, 1974). A fairly good fit was obtained to measured values with a very considerable investment of instrumentation and computer time.

Recent models have generally attempted to simplify the problem, by ignoring some interactions. In most contexts little downslope flow is thought to occur in the unsaturated phase, so that flow can be modelled as vertical unsaturated flow and lateral saturated flows, both overland and subsurface. If the downslope flows are treated as a single layer, as in the St. Venant equation for overland flow and eqn. (13) above for subsurface flow, then the problem is considerably reduced to a series of coupled one-dimensional equations. The bulk of the computation is then concerned with following infiltration for each element downslope. Some of these simplifications have been introduced into recent models, like the SHE model (Beven et al., 1980) and the IHDM model (Morris, 1980). The model of Jayarwardena and White (1977, 1979) perhaps come closest to the simplified specification proposed here, though further simplifying the infiltration component.

Another type of model which has developed explicitly from hillslope

hydrology is the partial or contributing area model, or variable source area model for human catchments. In analysing rainfall-runoff data, Betson (1964) defined contributing area as (peak stream discharge) $\div$ (peak rainfall intensity). He found that, for most eastern American catchments, the contributing area was a small fraction of the total catchment. In a forecasting sense, the contributing area has generally been associated with an area which is either directly saturated by return flow, or becomes saturated with the addition of some storm rainfall. Using, for illustration, the equilibrium pattern of deficits defined by eqn. (17) above, some areas may already be saturated (where $q_0 s/a < i$), while at others storm rainfall is soon able to fill the shallow unsaturated zone (i.e. deficit D is small). Subsequent rainfall on this dynamic contributing area runs off as saturation overland flow, which is responsible for a major part of the flood hydrograph in many humid catchments. It is evident, both from eqn. (17) and from mapping by Dunne and others (Dunne and Black, 1970a,b; Dunne et al., 1975), that saturated areas occur in many catchments, particularly around concave stream head hollows, and that they expand and contract in area during rain storms as illustrated in Fig. 5.

Engman and Rogowski (1974) combined infiltration equations with mapped soil properties to forecast saturated areas for Hortonian overland flow as a basis for routing. Troendle (Troendle, 1985; Hewlett and Troendle, 1975) has developed a series of related models which use a two-dimensional numerical solution for saturated and unsaturated water movement along a flow strip. Where surface elements near the slope base become saturated, the saturated area is added to the effective channel area receiving direct precipitation and the remainder of the hillside upslope is redivided into elements, with their highest density close to the edge of the saturated area. The model is designed to give its most accurate forecasts for the margins of the saturated area, which then generates overland flow. It was assumed that no Hortonian overland flow occurred. The model gave good agreement for a 0.38 km^2 mountain catchment in West Virginia (Furness no. 5) subdivided into nineteen flow strips. Application to a 0.24 km^2 catchment near Athens, Georgia (Whitehall watershed) was less successful. Beven and Kirkby (1979) attempted to simplify the subsurface flow regime still further, using eqns. (14) and (16) above to forecast (spatially uniform) runoff within each flow strip. Analyses of flow strip topography and soil properties are used to convert runoff to saturation deficit, giving saturated areas directly. Streamflow is then forecast from subsurface and overland flow contributions. This simplified model (TOPMODEL) was able to give moderately good forecasts over both storm and inter-storm periods but appeared to be least satisfactory in areas where saturated hollows were least evident in the field.

Since the early 1970s, there has been an explosion of models which explicitly include hillslope hydrological processes. From a position in which all relevant processes were included in a distributed form, there has been some retreat

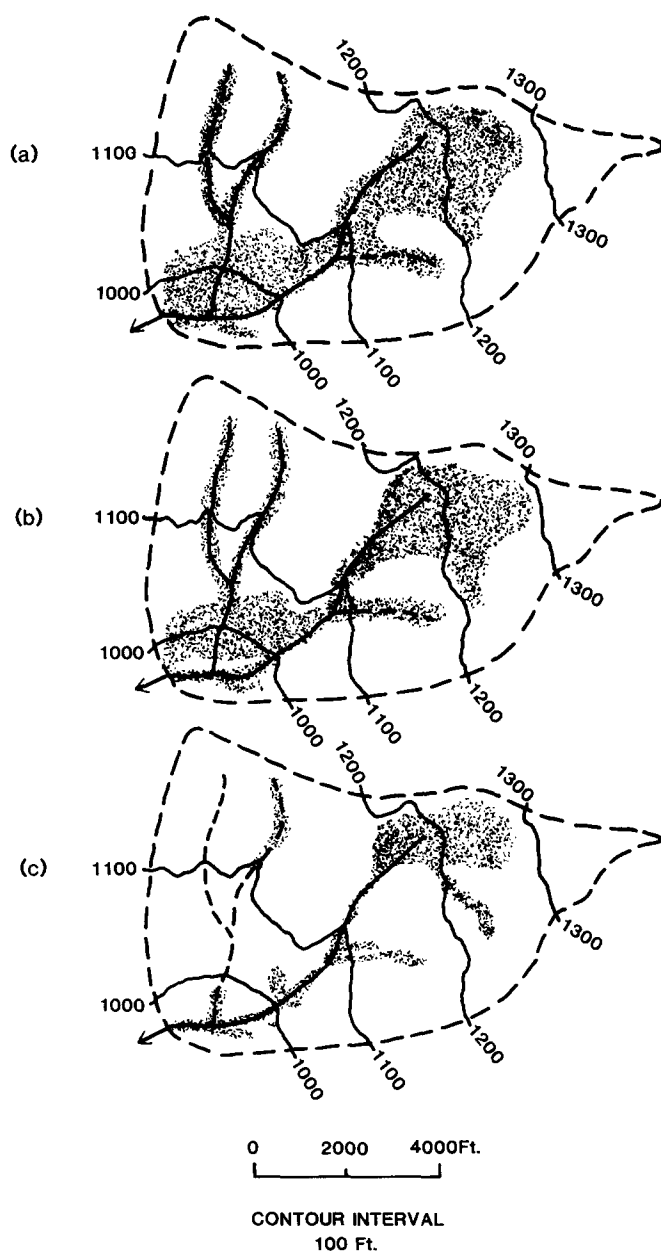


Fig. 5. Variation in the saturated contributing area for a small catchment in glacial till and schist near Randboro, Quebec (from Dunne et al., 1975). (a) 22 March 1973: 36% saturated; (b) 26 March 1973: 24% saturated; (c) 26 April 1973: 12% saturated.

towards models which are less complete, neglecting some processes and making less demands on both field data and computing resources, even though computer power is increasingly cheap and widespread. It is seen that such

models are more comprehensible and give forecasts which are no less reliable. The immediate future seems likely to continue this trend, perhaps towards reliable models which can run on desk top microcomputers as a routine management tool for individual catchments. At the same time there has been too strong a concentration on models for humid temperate areas, in which the Hortonian overland flow paradigm has given place to a saturation overland flow paradigm. The variety in hillslope hydrological response which is now being documented should lead to models which take account of a more complex interaction between overland flow and infiltration and of bypass and seepage pipe networks.

RELATIONSHIP TO THE CATCHMENT AS A WHOLE

Pursuing the logic of the previous section, flow strips should be combined to form whole catchments, at an acceptable level of aggregation on the basis of homogeneous response. Additional features needed are an effective channel routing algorithm and a good knowledge of rainfall distribution over the catchment. Both of these factors significantly influence the timing of catchment response, increasingly so in large catchments. Nevertheless, large catchments become progressively more linear in their response with increasing catchment area and it is both impractical and uneconomic to subdivide large catchments at the density of flow strips used for small catchments, where individual strips have areas of 1–10 ha.

Catchments may be thought of as a sequence of moisture stores, some in series and some in parallel. Thus, for example, flow from a point on the watershed to the stream outlet may pass through stores for surface detention, infiltration, unsaturated vertical percolation, saturated downslope flow and channel flow. For any such sequence of storage in series, the behaviour is dominated by that of the stores with the longest residence times: in most cases, the two slowest stores are enough to make a satisfactory model of the system as a whole. For such a case, the residence times on the hillside are of the order of:

Surface detention	0.1–1 h
Infiltration	1–20 h
Percolation	1–50 h
Downslope flow	1–12 h
Channel flow	0.5 h (1 km ²), 7 h (100 km ²), 100 h (10 ⁴ km ²)

Thus, for a small catchment, it may be efficient to make a model which takes account of infiltration, percolation and downslope flow. If the same catchment is, however, embedded in a large catchment, it may be preferable to simplify the hillslope response to take account of unsaturated percolation alone and to combine it with a channel routing procedure. There is therefore some justification in treating hillslope hydrology in a simplified manner within large catchments.

The comparative success of early unit hydrograph models, particularly in

catchments of over 100 km², also shows that much of the local variety and nonlinearity of individual hillslopes can be ignored or averaged within large catchments. The combination of hillslope outflows, routing them through the catchment, is greatly simplified if it can be assumed that kinematic wave velocities are spatially uniform at any moment for the catchment, although numerical solutions may be obtained with more general assumptions. The assumption of uniform velocities has been made by Surkan (1974) and Rodriguez-Iturbe and Valdez (1979), in his Geomorphological Unit Hydrograph concept. Beven (1979) has shown that channel kinematic wave velocities may be approximately uniform over time, even though average velocities depend on discharge. With these assumptions, catchment response becomes progressively more linear for large catchments (Kirkby, 1976), as channel travel times increasingly dominate the hydrograph form. In the Geomorphological Unit Hydrograph, the hillslope is treated as a single linear store with a constant characteristic response time throughout a catchment. The flow is then routed through the catchment, assuming a spatially constant kinematic wave velocity. Although this approach is too great a generalisation for a practical forecasting model, there is scope for using somewhat simplified hillslope models within a large basin context.

For example eqns. (14) and (16) provide a runoff and saturated area model for subcatchments, which is better validated than the simple linear store. For the large catchment, overland flow may be treated as instantaneous, so that hillslope routing is not required. There are consistent changes in hillslope form within a catchment documented by, among others, Carter and Chorley (1961) and Arnett (1971). Typically average hillside gradients increase downstream until about fifth-order streams and then decline for higher orders. This pattern may either be generalised, or documented for particular catchments, to forecast saturated areas which are more important in headwaters (in stream head hollows) and downstream (around flood plains) than in between. Though this example is not intended to be definitive, there is considerable scope for development of large catchment models which make use of hillslope data and hillslope hydrological processes to produce improved forecasting of only modest complexity.

A critical problem in combining hillslope and channel network processes into a hydrological model for a whole catchment is that there is no unambiguous method for determining the exact position of channel heads. Not only are different criteria used on published maps, but channel heads are often poorly visible from remotely sensed data (especially in forest) and may be indefinite even on the ground. If too low a density of channels is assumed in a forecast, then most existing hillslope models forecast increased attenuation and delay of hydrograph peaks. A satisfactory hillslope hydrological model must therefore be insensitive to the exact density of channels chosen. To achieve this, forecasts from the base of slopes and from the smallest channels must form a continuum of response. Given present knowledge of hillslope processes, response in these transitional areas must therefore be represented

by overland, pipe or channel flow, even for humid areas. Whether or not this difficulty is an artefact of the modelling process, a better understanding of non-Darcian elements of hillslope drainage may demonstrate that the continuum between the base of slopes and small channels is a physical reality.

CONCLUSION

In the 1960s and 1970s, hillslope hydrology developed as a distinct topic through the application of new field observations to develop a generation of physically based forecasting models. These took account of unsaturated and saturated flow within the regolith, and used the growth of more widespread computer power to produce numerical solutions to previously intractable problems. The great improvements which were made in forecasting hillslope and small catchment storm hydrographs, particularly for humid areas, can now be recognised. With hindsight, the most complete and distributed models showed little forecasting advantage over simpler approaches; and this is true for large as well as small catchments.

Since the mid 1970s, models incorporating subsurface flow have been applied to a wider range of environments. Just as Hortonian overland flow had previously been rejected as a universal process, it appears that additional processes must now also be recognised, and incorporated into the next generation of models, even if initially with a less secure physical and mathematical basis than the Richards equation. Most notably, a richer range of hillslope flow processes must be included, in response to the diversity of climates, and of soils within and between slopes. In particular, recent work should focus attention on the role of spatial variability in surface and soil properties and on the role of bypass and seepage macropipes and pipes. Surface properties are important in influencing the spatial distribution of infiltration and overland flow. Subsurface properties, particularly the existence of lines of preferential flow including macropores and pipes, influence the total response of a hillslope and call into question whether it can be treated as a Darcian flow system. Pipe systems pose a particular challenge to forecasters, since their density, or indeed their presence, can only be sampled sparsely for most forecasting applications.

The growing availability of remotely sensed data also poses a challenge for hillslope hydrology. Although the physical arguments favour an understanding of hillslope hydrology on the basis of flow strips, remotely sensed data are generally available for grid squares. How are hillslope hydrologists best able to take advantage of these data to increase forecasting power; how indeed can hillslope hydrologists produce forecasts for grid squares, to link with atmospheric processes in particular? Notwithstanding the substantial amount of current work devoted to these issues, the surface and subsurface hydrology remain relatively intractable to date.

Hillslope hydrology is an essential and active growth area, which has benefitted hugely from the interaction between fieldwork and modelling. In its

short history, theory has continually been overturned by field observation, and this creative relationship, involving engineers, field scientists and mathematicians still has a great deal to contribute.

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